

C5001: Internet Concept & Web Design

Tier 1

Question	Unit	Marks	Years	Repeat
What do you mean by Internet? What are the applications and advantages of internet?	U1	10	2016–19, 2019–22, 2020–23	1
Define TCP/IP model in detail. Discuss its layers and functions. Explain different protocols of each layer with suitable examples.	U2	10	2019–22, 2022–25	2
Create a HTML page with following features : Table, Links, Ordered List and unordered list.	U2	10	2015–18, 2016–19	2
Write about the html and XML. What are the difference between HTML and XML. Also write about XML and html programs for displaying any message.	U2/U3	10	2020–23	1
What are the advantages of javascript ? What are the main purpose of using javascript on the websites.	U2	10	2020–23	1
What is javascript operator ? Write down about all javascript operators.	U2	10	2019–22	1
What is Java Script Arrays ? Explain, with example, how it is created and accessed.	U2	10	2016–19	1
What is JSP ? What are the JSP Life Cycle Phases and JSP Life Cycle Methods ? Explain.	U4	10	2016–19, 2019–22	2
Write about jsp and what are the different types of jsp tags, discuss all jsp tags with examples.	U4	10	2020–23	1
What is PHP ? What are its features and applications ? Discuss about loops of the PHP.	U5	10	2020–23	1
Discuss about PHP, features of PHP and uses of PHP. What is string and write about some predefined string handling functions of PHP.	U5	10	2019–22	1

Tier 2

Question	Unit	Marks	Years	Repeat
Write about Internet Domains, Domain registration, Transmission Control Protocol.	U1	10	2020–23	1
Explain how an internet server establishes connectivity and verify its identity on the internet.	U1	10	2022–25	1
What is javascript ? And discuss about javascript function with a program which performs addition operation between two given values.	U2	10	2019–22	1
What is data types ? What are the different data types supported in javascript ? Write a program in javascript for displaying your name with a program.	U2	10	2020–23	1
What do you mean by validation? Using Java Script validate an input field that contains email address.	U2	10	2015–18	1
What is markup language and what is the relationship between XML, HTML and DTD?	U3	10	2015–18	1
Why XML is important? How are SGML, HTML & XML related?	U3	10	2016–19	1
What is DTD? Also write about internal DTD and external DTD with examples.	U3	10	2019–22	1

Tier 3

Question	Unit	Marks	Years	Repeat
What is TCP / IP ?	U1	3	2015–18, 2020–23	2
How internet is useful in education ?	U1	3	2020–23	1
Write about web browser.	U1	3	2020–23	1
What is tag and attribute in html?	U2	3	2020–23	1
What is CSS ?	U2	3	2019–22	1
What do you mean by type casting in javascript?	U2	3	2020–23	1
What is page directive in jsp ?	U4	3	2020–23	1
Write about some string functions of PHP.	U5	3	2020–23	1

C5002: Design & Analysis of Algorithms

Tier 1

Question	Unit	Marks	Years	Repeat
Write the asymptotic notations used for best case, average case and worst case analysis of an algorithm.	U1	10	2015–18, 2016–19, 2020–23	3
What is an algorithm ? Explain the different characteristics of an algorithm.	U1	10	2016–19, 2019–22	2
Explain detail about Merge Sort. Illustrate the algorithm with a numeric example. Prove its time complexity.	U2	10	2016–19, 2020–23	2
Write the complete algorithm for Quicksort. Calculate its complexity in worst, best and average cases.	U2	10	2015–18, 2016–19, 2019–22	3
Define spanning tree. Write Kruskal's algorithm for finding minimum cost spanning tree.	U3	10	2020–23	1
Explain Greedy criterion. Write a Prim's algorithm to find minimum cost spanning tree.	U3	10	2016–19, 2019–22	2
Solve the following recurrence using the Master method.	U4	10	2019–22, 2020–23	2
Solve the following Recurrence using Recursion tree method: $T(n) = 2T(n/4) + 2T(n/10) + 1$.	U4	10	2019–22	1
Trace how Depth First Search (DFS) traverses a given graph.	U5	10	2015–18, 2016–19, 2019–22, 2020–23	4

Tier 2

Question	Unit	Marks	Years	Repeat
What is Divide and Conquer technique ? Explain Max-Min algorithm and analyse.	U2	10	2020–23	1
Compare the various programming paradigms such as divide-and-conquer, dynamic programming and greedy approach.	U2	10	2020–23	1
Design a recursive algorithm for binary search and analyze its efficiency.	U2	10	2015–18, 2016–19, 2019–22	3
Define Hashing Technique with the help of an example.	U5	10	2015–18, 2016–19, 2019–22, 2020–23	4

Tier 3

Question	Unit	Marks	Years	Repeat
Write time complexities of Binary search and Merge sort.	U1/U2	3	2019–22	1
Mention Best Case, Average Case and Worst Case complexities of linear Search.	U1	3	2019–22	1
Arrange following complexities in ascending order $O(n \log n)$, $O(1)$, $O(n^3)$, $O(n^2)$, $O(1)$.	U1	3	2019–22	1
What are the applications of BFS?	U5	3	2020–23	1
Describe how Kruskal's algorithm is different from Prim's algorithm.	U3	3	2015–18, 2019–22, 2020–23	3